20245227 이찬호

9-14. 선형제어시스템의 특성방정식이 다음과 같다. $K \geq 0$에서 근궤적을 그려라.

(a) $s^3+3s^2+(K+2)s+5K=0$
(b) $s^3+s^2+(K+2)s+3K=0$
(c) $s^3+5Ks^2+10=0$
(d) $s^4+(K+3)s^3+(K+1)s^2+(2K+5)+10=0$
(e) $s^3+2s^2+2s+K(s^2-1)(s+2)=0$
(f) $s^3-2s+K(s+4)(s+1)=0$
(g) $s^4+6s^3+9s^2+K(s^2+4s+5)=0$
(h) $s^3+2s^2+2s+K(s^2-2)(s+4)=0$
(i) $s(s^2-1)+K(s+2)(s+0.5)=0$
(j) $s^4+2s^3+2s^2+2Ks+5K=0$
(k) $s^5+2s^4+3s^3+2s^2+s+K=0$

: 근궤적 방법

(a) $s^3+3s^2+(k+2)s+5k=0$

$\Rightarrow s^3+3s^2+2s+(s+5)k=0$

$\Rightarrow s(s+1)(s+2)+(s+5)k=0$

1) $1+k\frac{N(s)}{D(s)}=0$ 만들기

$\Rightarrow 1+k\frac{(s+5)}{s(s+1)(s+2)}=0$

영점: -5
극점: $0,-1,-2$

2) 점근선 각도 구하기

n = 극점수 m = 영점수

$\theta=\frac{(2i+1)}{n-m}\times 180^\circ=\frac{2i+1}{2}\times 180^\circ \quad (i=0,1,2\cdots,n-m-1)$

$\theta_0=90^\circ$, $\theta_1=270^\circ$

3) 점근선이 실수축과 만나는 위치 구하기

$\sigma=\frac{\sum 극점-\sum 영점}{n-m}=\frac{(0-1-2)-(-5)}{3-1}=1$

4) 이탈점 구하기 →←

$k=-\frac{s(s+1)(s+2)}{(s+5)}$ ★ $\frac{dk}{ds}=0$ $\left(\frac{a}{b}\right)'=\frac{a'b-ab'}{b^2}$

$=-\frac{s^3+3s^2+2s}{s+5}$

$\frac{dk}{ds}=-\frac{(3s^2+6s+2)(s+5)-(s^3+3s^2+2s)}{(s+5)^2}$

★ 분모 $\neq 0$ $(s\neq -5)$
분자 $=0$

$s=-6.9443, -1.609, -0.448$

불가능

5) 허수축 통과 구간 구하기

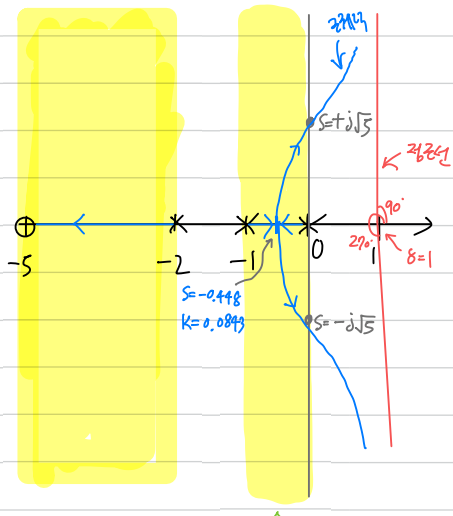
$s^3+3s^2+(k+2)s+5k=0$을 Routh 표

$a \quad b$
$c \quad d$
$\frac{cb-ad}{c}$

s^3	1	$k+2$
s^2	3	$5k$
s^1	$\frac{3(k+2)-1\times 5k}{3}$	0
s^0	$5k$	

2) $3s^2+5k=0$ $3s^2+15=0$ $s^2=\pm j\sqrt{5}$

1) $\frac{6-2k}{3}=0 \Rightarrow k=3$



근궤적의 영점, 극점의 갯수와 홀수인 구간만 클래

가능

(근궤적 구간에 있고 $K\geq 0$ 이어야함)

근궤적구간의 s를 $K=\frac{s\cdots}{s\cdots}$ 식에 대입해서

(b) $s^3+s^2+(k+2)s+3k=0$

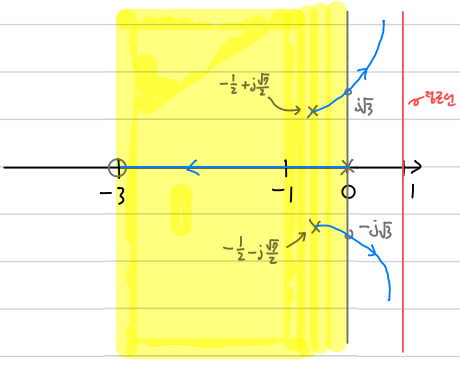
$\Rightarrow s(s^2+s+2)+k(s+3)=0$

$$1+k\frac{s+3}{s(s^2+s+2)}=0$$

영점: -3

극점: 0, $-\frac{1}{2}+j\frac{\sqrt{7}}{2}$, $-\frac{1}{2}-j\frac{\sqrt{7}}{2}$

$s=\frac{-1\pm\sqrt{1-8}}{2}=\frac{-1\pm j\sqrt{7}}{2}$



- 점근선 각도

$n=3$ $m=1$

$\theta=\frac{(2i+1)}{2}\times 180^\circ$ $\theta_0=90^\circ$, $\theta_1=270^\circ$

- 점근선이 실수축과 만나는 점

$\sigma=\frac{\sum \text{극점}-\sum \text{영점}}{n-m}=\frac{-1-(-3)}{2}=1$

- 이탈점

$$k=-\frac{s^3+s^2+2s}{s+3} \qquad \frac{dk}{ds}=\frac{(3s^2+2s+2)(s+3)-(s^3+s^2+2s)}{(s+3)^2}=\frac{2s^3+10s^2+6s+6}{(s+3)^2}$$

$s^3+5s^2+3s+3=0$

$s=-4.480$, $-0.260\pm j0.776$ 즉, 실수축 이탈점 X

근궤적 범위 X 허수축서 X

허수축 통과점 찾기

$s^3+s^2+(k+2)s+3k=0$

s^3	1	$k+2$
s^2	1	$3k$
s^1	$\frac{k+2-3k}{1}$	0
s^0	$3k$	

$s^2+3k=0 \Rightarrow s^2+3=0$ $s=\pm j\sqrt{3}$

$-2k+2=0 \Rightarrow k=1$

(c) $S^3+5KS^2+10=0$

$\Rightarrow S^3+10+K(5S^2)=0$

$\Rightarrow 1+K\dfrac{5S^2}{S^3+10}=0$ 영점: 0 (중근) m=2

극점: −2.154, 1.077±j1.866 n=3

◦ 점근선 각도

$\theta=\dfrac{2i+1}{n-m}\times 180^\circ$ $\theta_0=180^\circ$

◦ 점근선이 실수 축과 만나는 위치

$\delta=\dfrac{\Sigma 극점-\Sigma 영점}{n-m}=0$

◦ 이탈점 구하기

$K=-\dfrac{S^3+10}{5S^2}$ $\dfrac{dK}{dS}=\dfrac{3S^2(5S^2)-(S^3+10)10S}{(5S^2)^2}=-\dfrac{5S^4-100S}{25S^4}$ $(S\neq 0)$

$5S(S^3-20)=0$ $S=0$, 2.714, −1.357±j2.351

× 근궤적 범위 X

즉 실수축 이탈점이 없음

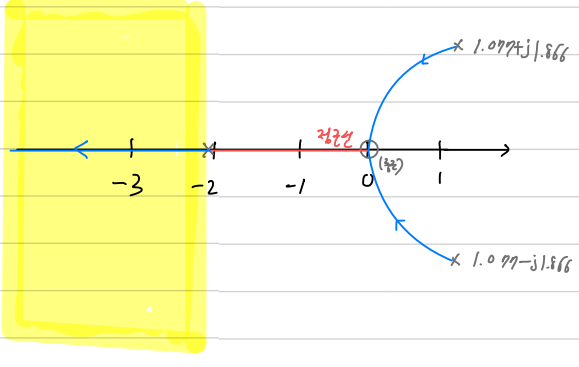
• 허수축 통과점

$S^3+5KS^2+0S+10=0$

S^3	1	0
S^2	5k	10
S^1	$-\frac{10}{5k}$	0
S^0	10	

$-\frac{2}{k}=0$ (k>0 에서 0인값이 없음)

즉, k≥0 에서 허수축 통과 X



(d) $s^4+(K+3)s^3+(K+1)s^2+(2K+5)+10=0$

$\Rightarrow s^4+3s^3+s^2+15+K(s^3+s^2+2)=0$

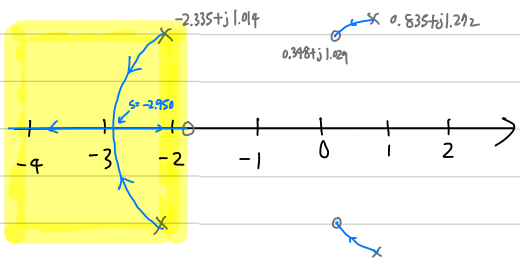
$1+K\dfrac{s^3+s^2+2}{s^4+3s^3+s^2+15}=0$

영점 : $-1.696,\ 0.348\pm j1.029$

극점 : $-2.335\pm j1.014,\ 0.835\pm j1.292$

• 점근선 각도

$\theta=\dfrac{2i+1}{n-m}\times 180^\circ \quad \theta_0=180^\circ$



· 점근선의 실수축과 만나는 위치

$\sigma=\dfrac{\sum 극점-\sum 영점}{n-m}=\dfrac{-3-(-1)}{1}=-2$

·이탈점

$K=-\dfrac{s^4+3s^3+s^2+15}{s^3+s^2+2} \qquad \dfrac{dk}{ds}=-\dfrac{(4s^3+9s^2+2s)(s^3+s^2+2)-(s^4+3s^3+s^2+15)(3s^2+2s)}{(s^3+s^2+2)^2}$

$s(s^5+2s^4+2s^3+8s^2-27s-26)=0$

$s=\underset{\text{가능}}{-2.95},\ -0.795,\ 0,\ 1.996$

• 허수축 통과점

$s^4+(K+3)s^3+(K+1)s^2+0s+(2K+15)=0$

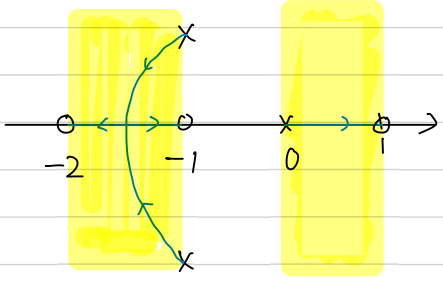
s^4	1	$K+1$	$2K+15$
s^3	$K+3$	0	0
s^2	$K+1$	$2K+15$	0
s^1	$-\frac{(K+3)(2K+15)}{K+1}$	0	
s^0	$2K+15$		

허수축 교차점 없음

(e) $s^3+2s^2+2s+k(s^2-1)(s+2)=0$

$$1+k\frac{(s-1)(s+1)(s+2)}{s(s^2+2s+2)}=0$$

영점: 1, -1, -2
극점: 0, $-1\pm j$



• 점근선 개수

$n-m=3-3=0$ (점근선 없음)

• 이탈점

$$K=-\frac{s^3+2s^2+2s}{(s^2-1)(s+2)}=-\frac{s^3+2s^2+2s}{s^3+2s^2-s-2}$$

$$\frac{dK}{ds}=\frac{(3s^2+4s+2)(s^3+2s^2-s-2)-(s^3+2s^2+2s)(3s^2+4s-1)}{(s^3+2s^2-s-2)^2}$$

$$=\frac{6s^3+12s^2+8s+4}{(s^3+2s^2-s-2)^2}$$

$3s^3+6s^2+4s+2=0$

$s=-1.3848, -0.3076\pm j0.6219$

가능

• 허수축 통과점

$(1+k)s^3+2(1+k)s^2+(2-k)s-2k=0$

s^3	$1+k$	$2-k$	
s^2	$2(1+k)$	$-2k$	
s^1	2	0	$\frac{2(1+k)(2-k)-(1+k)(-2k)}{2(1+k)}$
s^0	$-2k$		

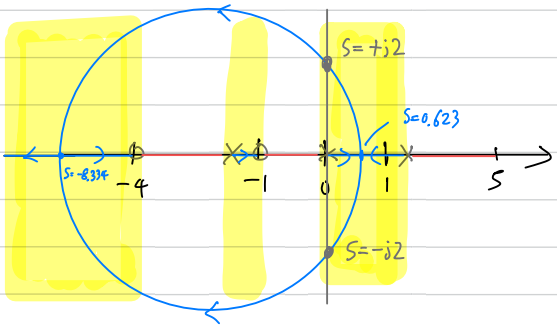
허수축 통과점 X

(f) $s^3-2s+k(s+4)(s+1)=0$

$$1+K\frac{(s+4)(s+1)}{s(s-\sqrt{2})(s+\sqrt{2})}=0$$

영점 : $-4, -1$

극점 : $-\sqrt{2}, 0, \sqrt{2}$



· 근궤적 각도

$n-m=1$ 이므로 $\theta=180°$

· 근궤적 중심

$$\sigma=\frac{\Sigma극점-\Sigma영점}{n-m}=5$$

· 이탈점 구하기

$$K=-\frac{s^3-2s}{(s+4)(s+1)}=-\frac{s^3-2s}{s^2+5s+4}$$

$$\frac{dK}{ds}=-\frac{s^4+10s^3+14s^2-8}{(s^2+5s+4)^2} \quad \text{or} \quad s=0.6229, -8.3339, -1.1445\pm j0.4808$$

가능

· 허수축 통과점

$s^3+ks^2+(5k-2)s+4k=0$

s^3	1	$5k-2$	
s^2	k	$4k$	$ks^2+4k=0$
s^1	$5k-6$	0	$k=1.2$
s^0	$4k$		

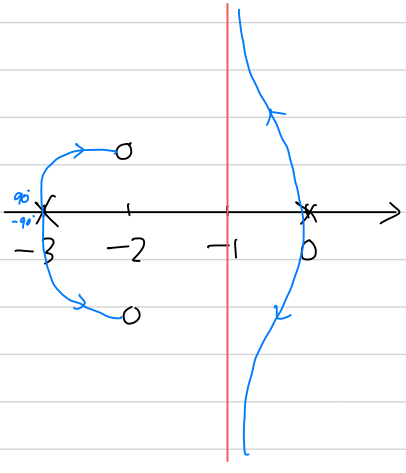
$s=\pm j2$

(9) $S^4+6S^3+9S^2+K(S^2+4S+5)=0$

$$1+K\frac{S^2+4S+5}{S^2(S+3)^2}=0$$

영점: $-2\pm j$

극점: 0(중근), -3(중근)



실수축 위에 근궤적 X

· 점근선 각도 $n=4$ $m=2$

$$\theta=\frac{2i+1}{2}\times 180^\circ \quad \theta_0=90^\circ,\ \theta_1=270^\circ$$

· 점근선 중심

$$\sigma=\frac{\Sigma극점-\Sigma영점}{n-m}=-1$$

· 이탈점

$$K=-\frac{S^2(S+3)^2}{S^2+4S+5} \qquad \frac{dK}{dS}=-\frac{2S(S+3)(S^3+6S^2+10S+15)}{(S^2+4S+5)^2}$$

$S=0$, $S=-3$ $S^3+6S^2+10S+15=0$ $S=-1.9537$ (대입하면 $K=-4.5038$ 이므로 X)

허수축 통과점

$S^4+6S^3+(9+K)S^2+4KS+5K=0$

S^4	1	$9+K$	$5K$
S^3	6	$4K$	0
S^2	$\frac{K+27}{3}$	$5K$	0
S^1	$\frac{2K(2K+9)}{K+27}$	0	
S^0	$5K$		

영점 $-2+j$ 에서 원점을 보는 각도 $(-2,1)(0,0)$ $\tan^{-1}\frac{1}{2}=26.565^\circ$

영점 $-2-j$ 에서 원점을 보는 각도 $(-2,-1)(0,0)$ $\theta=-26.565$ ⓐ $=0^\circ$

극점 -3(중근)에서 원점으로 $0^\circ+0^\circ=0^\circ$

극점 0에서 0은 θ° $\theta+\theta=2\theta$

영점 각도합 − 극점 각도합 $=(2i+1)\times 180^\circ$ $-2\theta=-180^\circ$ $\theta=90^\circ$

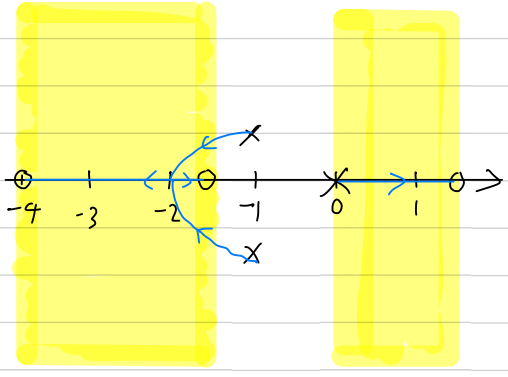
$-2\theta=180^\circ$ $\theta=-90^\circ$

(h) $s^3+2s^2+2s+K(s^2-2)(s+4)=0$

$$1+K\frac{(s-\sqrt{2})(s+\sqrt{2})(s+4)}{s(s^2+2s+2)}=0$$

영점: $-4, \pm\sqrt{2}$

극점: $0, -1\pm j$



- 점근선 각도

$n=3$ $m=3$, $n-m=0$ 점근선 X

- 이탈점, 합류점

$$K=-\frac{s^3+2s^2+2s}{(s^2-2)(s+4)}=-\frac{s^3+2s^2+2s}{s^3+4s^2-2s-8}$$

$$\frac{dk}{ds}=-\frac{(3s^2+4s+2)(s^3+4s^2-2s-8)-(s^3+2s^2+2s)(3s^2+8s-2)}{(s^3+4s^2-2s-8)^2}$$

$$=-\frac{2s^4+8s^3+36s^2+32s+16}{(s^3+4s^2-2s-8)^2}$$

$s^4+4s^3+18s^2+16s+8=0$

$s=-2$ (가능), -0.9466

- 출발각 구하기

극점 $-1+j$에서 출발하는 방향

- 영점 -4에서 $\theta=\tan^{-1}\frac{1}{3}=18.435^\circ$
- 영점 $\sqrt{2}$에서 $\theta=\tan^{-1}\frac{1}{1+\sqrt{2}}=157.5^\circ$
- 영점 $-\sqrt{2}$에서 $\theta=\tan^{-1}\frac{1}{\sqrt{2}-1}=67.5^\circ$
- 극점 0에서 $\theta=\tan^{-1}\frac{1}{1}=45^\circ$
- 극점 $-1-j$에서 $\theta=90^\circ$

영점각의합 - 다른극점합 $-\theta=(2i+1)180^\circ$

$(18.435+67.5+157.5)-(135+90^\circ)-\theta=-180^\circ$

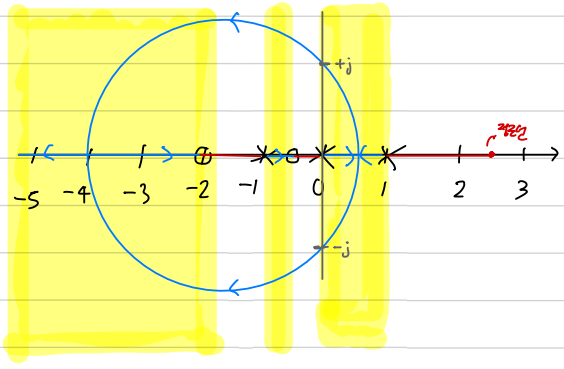
$\theta=198.435$ (위쪽극점에서 왼쪽여재)

아래 극점은 대칭이라

$\theta=360^\circ-198.435=161.565^\circ$

(i) $S(S^2-1)+K(S+2)(S+0.5)=0$

$1+K\dfrac{(S+2)(S+0.5)}{S(S+1)(S-1)}=0$ 영점: $-2, -0.5$ 극점: $-1, 0, 1$



• 점근선 각도

$n=3, m=2$

$\theta=\dfrac{(2i+1)}{n-m}\times 180^\circ$ $\theta_0=180^\circ$

• 점근선 중심

$\sigma=\dfrac{\Sigma극점-\Sigma영점}{n-m}=\dfrac{(-1+0+1)-(-2-0.5)}{1}=2.5$

• 이탈점, 합류점

$K=-\dfrac{S(S^2-1)}{(S+2)(S+0.5)}=-\dfrac{S^3-S}{S^2+2.5S+1}$

$\dfrac{dk}{ds}=-\dfrac{(3S^2-1)(S^2+2.5S+1)-(S^3-S)(2S+2.5)}{(S^2+2.5S+1)^2}$ $S^4+5S^3+4S^2-1=0$

$S=0.40245, -4.02048, -0.69098\pm j0.37494$

가능 가능

• 허수축 통과 점

$S^3+KS^2+(2.5K-1)S+K=0$

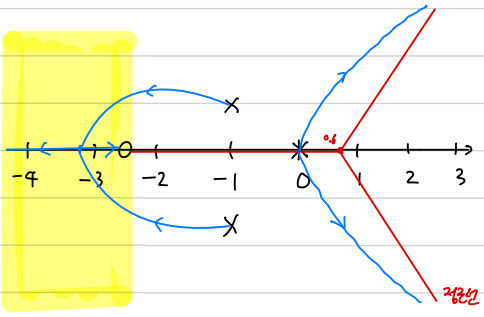
S^3	1	$2.5K-1$	
S^2	K	K	$KS^2+K=0$ $S^2=-1$ $S=\pm j$
S^1	$\dfrac{K(2.5K-1)-K}{K}$	0	$2.5K-2=0$ $K=0.8$
S^0	K		

(j) $S^4+2S^3+2S^2+2KS+5K=0$

$$1+K\frac{2S+5}{S^2(S^2+2S+2)}=0$$

영점: -2.5

극점: 0 (중근), $-1\pm j$



- 점근선 각도

$n=4$ $m=1$

$\theta=\frac{(2i+1)}{3}\times 180^\circ$ $\theta_0=60^\circ$, $\theta_1=180^\circ$, $\theta_2=300^\circ$

- 점근선 중심

$$\sigma=\frac{0+0+(-1+j)+(-1-j)-(-2.5)}{3}=\frac{1}{6}$$

- 이탈점, 합류점

$$K=-\frac{S^4+2S^3+2S^2}{2S+5} \qquad \frac{dK}{dS}=-\frac{(4S^3+6S^2+4S)(2S+5)-2(S^4+2S^3+2S^2)}{(2S+5)^2}$$

$$2S(3S^3+14S^2+17S+10)=0$$

$S=0$, -3.232, -0.768

가능

- 출발각 구하기

- $S=0$ (중근)

영점 -2.5 에서 $\theta=0^\circ$

극점 $-1+j$ 에서 $\theta=-\tan^{-1}\frac{1}{1}=-45^\circ$

극점 $-1-j$ 에서 $\theta=\tan^{-1}\frac{1}{1}=45^\circ$

$$0-(-45)-45-2\theta=(2i+1)\times 180^\circ$$

$$-2\theta=(2i+1)180^\circ \qquad \theta=90^\circ, 270^\circ$$

- $-1+j$

영점 -2.5 에서 $\theta=\tan^{-1}\frac{1}{1.5}=33.69^\circ$

극점 0 (중근)에서 $\theta=180-\tan^{-1}\frac{1}{1}=135^\circ$ (2개)

극점 $-1-j$ $\theta=90^\circ$

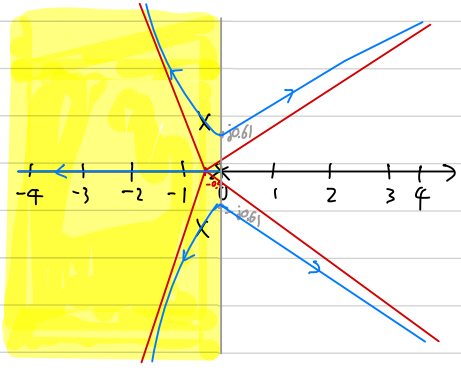
$$33.690-135-135-90-\theta=-180^\circ$$

$$\theta=-146.310^\circ$$

$\theta=360-146.310^\circ=213.690^\circ$

(k) $S^5+2S^4+3S^3+2S^2+S+K=0$

$1+K\dfrac{1}{S(S^2+S+1)^2}=0$ 영점: X 극점: 0, $-0.5+j0.866$(3중), $-0.5-j0.866$(3중)



• 점근선 각도

$n-m=5$

$\theta=\dfrac{2i+1}{5}\times 180^\circ$ $\theta_0=36^\circ, \theta_1=108^\circ, \theta_2=180^\circ, \theta_3=252^\circ, \theta_4=324^\circ$

• 점근선 중심

$\sigma=\dfrac{-2-0}{5}=-0.4$

• 이탈점, 합류점

$K=-S(S^2+S+1)^2$

$\dfrac{dK}{dS}=-(S^2+S+1)(5S^2+3S+1)$ $S=-\frac{1}{2}\pm j\frac{\sqrt{3}}{2}, \frac{-3\pm j\sqrt{11}}{10}$

*모두 복소수이므로 이탈점 X

• 중복극점의 출발각

$p=-\frac{1}{2}+j\frac{\sqrt{3}}{2}$

영점각의합 $=0^\circ$ 극점각은 $120^\circ, 90^\circ, 90^\circ$

$0^\circ-120^\circ-90^\circ-90^\circ-2\theta=(2i+1)\times 180^\circ$

$\theta=120^\circ, 300^\circ$ 아래쪽은 $\theta=240^\circ, 60^\circ$

• 허수축 통과점

S^5	1	3	1
S^4	2	2	K
S^3	2	K늘	0
S^2	1+K늘	K	0
S^1	$\frac{4-8K-K^2}{2(K+2)}$	0	
S^0	K		

$S=\pm j0.61803$

9-18. 단위피드백 제어시스템이 다음과 같은 전방경로 전달함수를 가진다. $K \geq 0$에서 근궤적선도를 그려라. 모든 이탈점에서 K의 값을 구하라.

(a) $G(s)=\dfrac{K}{s(s+10)(s+20)}$

(b) $G(s)=\dfrac{K}{s(s+1)(s+3)(s+5)}$

(c) $G(s)=\dfrac{K(s-0.5)}{(s-1)^2}$

(d) $G(s)=\dfrac{K}{(s+0.5)(s-1.5)}$

(e) $G(s)=\dfrac{K\left(s+\frac{1}{3}\right)(s+1)}{s\left(s+\frac{1}{2}\right)(s-1)}$

(f) $G(s)=\dfrac{K}{s(s^2+6s+25)}$

(a) $G(S)=\dfrac{K}{S(S+10)(S+20)}$

$1+G(S)=0 \quad 1+K\dfrac{1}{S(S+10)(S+20)}=0$ 영점: X 극점: 0, -10, -20

$n=3 \quad m=0$

• 점근선 각도

$\theta=\dfrac{(2i+1)\times 180^\circ}{3}$ $\theta_0=60^\circ,\ \theta_1=180^\circ,\ \theta_2=300^\circ$

• 점근선 중심

$\delta=\dfrac{0-10-20}{3}=-10$

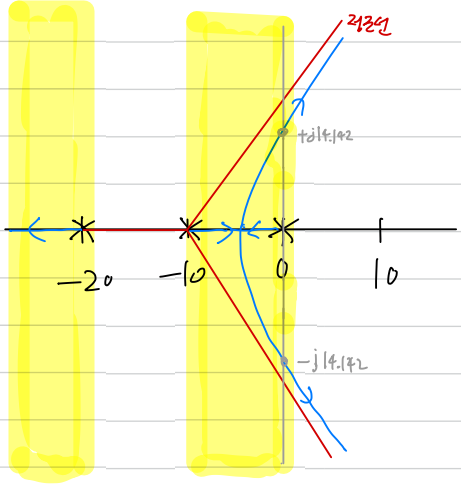
• 이탈점

$k=-S(S+10)(S+20)$ 대입

$\dfrac{dk}{ds}=-(3S^2+60S+200)$ $S=-10\pm\dfrac{10\sqrt{3}}{3}$

$=-4.2265,\ -15.7735$

가능



$S=-4.2265$ 일때 $K=384.9$

• 허수축 통과점

$S^3+30S^2+200S+K=0$

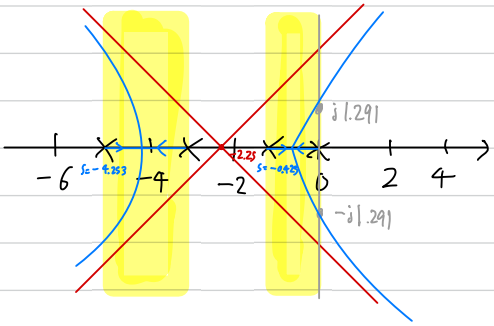
S^3	1	200	
S^2	30	K	$30S^2+K=0$ $S^2=-200$ $S=\pm j14.14$
S^1	$\frac{6000-K}{30}$	0	$6000-K=0$, $K=6000$
S^0	K		

답: $S=-4.2265$, $K=384.9$

(b) $G(s) = \dfrac{k}{s(s+1)(s+3)(s+5)}$

$1 + \dfrac{k}{s(s+1)(s+3)(s+5)} = 0$ 영점: X

극점: -5, -3, -1, 0



$n = 4 \quad m = 0$

· 점근선 개수

$n - m = 4$

$\theta = \dfrac{(2i+1) \times 180^\circ}{4}$ $\theta_0 = 45^\circ, \theta_1 = 135^\circ, \theta_2 = 225^\circ, \theta_3 = 315^\circ$

· 점근선 중심

$\sigma = \dfrac{-5-3-1+0}{4} = -2.25$

· 이탈점

$k = -s(s+1)(s+3)(s+5) = -(s^4 + 9s^3 + 23s^2 + 15s)$

$\dfrac{dk}{ds} = -(4s^3 + 27s^2 + 46s + 15)$ $4s^3 + 27s^2 + 46s + 15 = 0$

$s = \underbrace{-4.25375}_{\text{가능}}, -2.07046, \underbrace{-0.42579}_{\text{가능}}$ 대입

$s = -4.25375$ 일때 $k = 12.94945$

$s = -0.42579$ 일때 $k = 2.8789$

· 허수축 통과점

$s^4 + 9s^3 + 23s^2 + 15s + K = 0$

s^4	1	23	k
s^3	9	15	0
s^2	$\frac{64}{3}$	k	0
s^1	$\frac{960-27k}{64}$	0	
s^0	k		

$\frac{64}{3}s^2 + k = 0$ $s^2 = -\frac{3}{64}k$ $s = \pm j1.29$

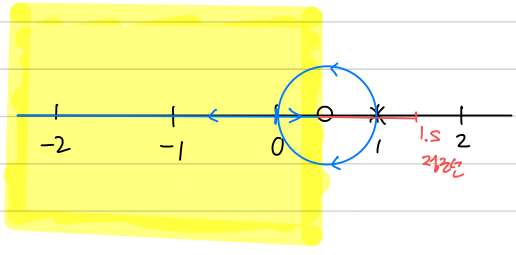
$960 - 27k = 0$ $k = 35.56$

답: $s = -4.25375, k = 12.94945$

$s = -0.42579, k = 2.8789$

(c) $G(s) = \dfrac{k(s-0.5)}{(s-1)^2}$

$1 + \dfrac{k(s-0.5)}{(s-1)^2} = 0$ 영점: 0.5 극점: 1(중근)



$n=2,\ m=1$

- 점근선 각도

$\theta = \dfrac{2i+1}{n-m} \times 180^\circ$ $\theta_0 = 180^\circ$

- 점근선 중심

$\sigma = \dfrac{(1+1)-0.5}{2-1} = 1.5$

- 이탈점, 합류점

$k = -\dfrac{(s-1)^2}{s-0.5}$ $\dfrac{dk}{ds} = -\dfrac{2(s-1)(s-0.5)-(s-1)^2}{(s-0.5)^2} = \dfrac{s(s-1)}{(s-0.5)^2}$

$s = 0, 1$ (가능) $s=0$ 일때 $K=2$

- 2중극점 $s=1$ 출발각

$0^\circ - 2\theta = (2i+1)180^\circ$

$-2\theta = -180^\circ$ $\theta = 90^\circ$

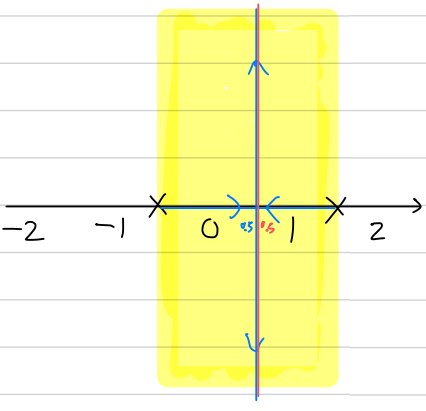
$-2\theta = 180^\circ$ $\theta = 270^\circ$

답: $s=0$, $k=2$

(d) $G(s)=\frac{K}{(s+0.5)(s-1.5)}$

$1+\frac{K}{(s+0.5)(s-1.5)}$ 영점: X 극점: $-0.5, 1.5$

$n=2$, $m=0$



- 점근선 갯수

$\theta=\frac{2+1}{2}\times 180^\circ$ $\theta=90^\circ, 270^\circ$

- 점근선 중심

$\sigma=\frac{-0.5+1.5}{2}=0.5$

- 이득점

$K=-(s+0.5)(s-1.5)$

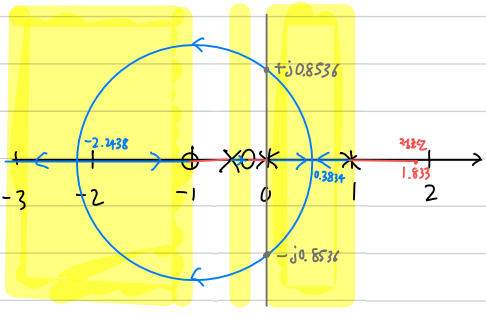
$\frac{dK}{ds}=-2s+1$ $s=0.5$, $K=1$

답: $s=0.5, K=1$

(e) $G(s)=\dfrac{k(s+\frac{1}{3})(s+1)}{s(s+\frac{1}{2})(s-1)}$

$1+\dfrac{k(s+\frac{1}{3})(s+1)}{s(s+\frac{1}{2})(s-1)}=0$ 영점: $-1, -\frac{1}{3}$ 극점: $-\frac{1}{2}, 0, 1$

$n=3$, $m=2$



· 점근선 각도

$\theta=\dfrac{2i+1}{1}\times 180^\circ$ $\theta=180^\circ$

· 점근선 중심

$\sigma=\dfrac{(-\frac{1}{2}+0+1)-(-1-\frac{1}{3})}{1}=1.833$

· 이탈점, 합류점

$k=-\dfrac{s(s+\frac{1}{2})(s-1)}{(s+\frac{1}{3})(s+1)}=-\dfrac{s^3-\frac{1}{2}s^2-\frac{1}{2}s}{s^2+\frac{4}{3}s+\frac{1}{3}}$

$\dfrac{dk}{ds}=-\dfrac{(3s^2-s-\frac{1}{2})(s^2+\frac{4}{3}s+\frac{1}{3})-(s^3-\frac{1}{2}s^2-\frac{1}{2}s)(2s+\frac{4}{3})}{(s^2+\frac{4}{3}s+\frac{1}{3})^2}=-\dfrac{s^4+\frac{8}{3}s^3+\frac{5}{6}s^2-\frac{1}{3}s-\frac{1}{6}}{(s^2+\frac{4}{3}s+\frac{1}{3})}$

미분

$6s^4+16s^3+5s^2-2s-1=0$

$s=-2.2438, 0.3834, -0.403\pm j0.1766$

가능

$s=-2.2438$, $k=5.34129$

$s=0.3834$, $k=0.21082$

· 허수축 통과점

$s^3+(k-\frac{1}{2})s^2+(\frac{4}{3}k-\frac{1}{2})s+\frac{k}{3}=0$

s^3	1	$\frac{4}{3}k-\frac{1}{2}$		실근이므로 X
s^2	$k-\frac{1}{2}$	$\frac{k}{3}$	$s^2=-\frac{k}{3(k-\frac{1}{2})}$	$s=\pm 0.47824, \pm j0.85865$
s^1	$\frac{(k-\frac{1}{2})(\frac{4}{3}k-\frac{1}{2})-\frac{k}{3}}{k-\frac{1}{2}}$	0	$k=0.20346, 0.92154$	0
s^0	$\frac{k}{3}$			

답: $s=-2.2438$, $k=5.34129$

$s=0.3834$, $k=0.21082$

(f) $G(s)=\dfrac{K}{s(s^2+6s+25)}$

$1+\dfrac{K}{s(s^2+6s+25)}=0$ 영점: X 극점: 0, $-3\pm j4$

$n=3,\ m=0$

◦ 점근선 각도

$\theta=\dfrac{2i+1}{3}\times 180^\circ$ $\theta_0=60^\circ,\ \theta_1=180^\circ,\ \theta_2=300^\circ$

◦ 점근선 중심

$\sigma=\dfrac{\sum 극점-\sum 영점}{n-m}=\dfrac{0+(-3+j4)+(-3-j4)}{3}=-2$

◦ 이탈점

$K=-(s^3+6s^2+25s)$

$\dfrac{dK}{ds}=-(3s^2+12s+25)$ $3s^2+12s+25=0$ $s=\dfrac{-12\pm\sqrt{12^2-4\cdot3\cdot25}}{6}=-2\pm j\dfrac{\sqrt{39}}{3}$

s가 모두 복소수 이므로 함수점, 이탈점 X

◦ 복소극점 출발각

· $-3+j4$

영점 각도 합 $=0^\circ$

극점 0 에서 $\theta=\tan^{-1}\frac{4}{3}=53.130^\circ$

$180^\circ-53.130^\circ=126.87^\circ$

극점 $-3-j4$에서 $\theta=90^\circ$

영점각도합 $-$ 극점 각도합 $-\theta=(2i+1)\times180^\circ$

$0-126.87^\circ-90^\circ-\theta=-180^\circ$

$\theta=-36.87^\circ=323.13^\circ$

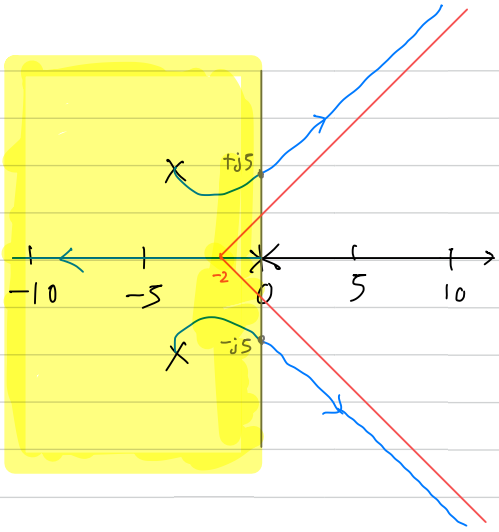
아래는 대칭이므로

$\theta=36.87^\circ$

· 허수축 통과점

$s^3+6s^2+25s+K=0$

s^3	1	25	$6s^2+K$ $6s^2+150=0$ $s=\pm j5$
s^2	6	K	
s^1	$\frac{150-K}{6}$	0	$\frac{150-K}{6}=0$ $K=150$
s^0	K		



9-22. 그림 7P-16의 제어시스템에서 $K = 100$일 때 특성방정식은 다음과 같다. $K_t \geq 0$에서 이 방정식의 근궤적을 그려라.

$$s^3+25s^2+(100K_t+2)s+100=0$$

$s^3+25s^2+(100k_t+2)s+100=0$

$\Rightarrow s^3+25s^2+2s+100+k_t(100s)=0$

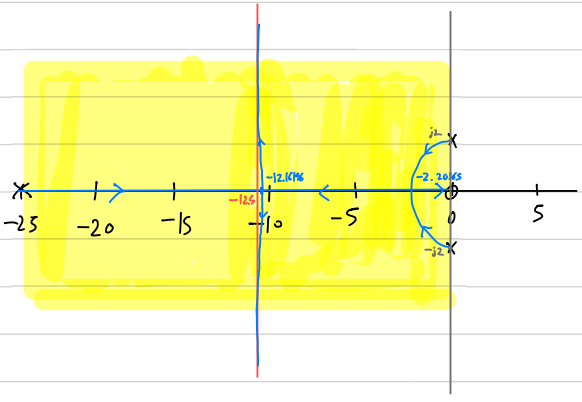
$$1+k_t\frac{100s}{s^3+25s^2+2s+100}=0$$

$s^3+25s^2+2s+100=0$

극점: $s=-25.07924,\ 0.03962\pm j1.99644$

영점: 0

$n=3,\ m=1$



- 점근선 각도

$\theta=\frac{2i+1}{2}\times 180^\circ$ $\theta_0=90^\circ,\ \theta_1=270^\circ$

- 점근선 중심

$\sigma=\frac{-25-0}{2}=-12.5$

- 분리점, 이탈점

$K_t=-\frac{s^3+25s^2+2s+100}{100s}$, $\frac{dk_t}{ds}=\frac{(3s^2+50s+2)(100s)-100(s^3+25s^2+2s+100)}{(100s)^2}$

$\frac{dk_t}{ds}=-\frac{2s^3+25s^2-100}{100s^2}$ $(s\neq 0)$

$2s^3+25s^2-100=0$

$s=-12.16196,\ -2.20365,\ 1.86562$

가능

- 복소극점 출발각

$p=0.03962+j1.99644$

영점 0 에서 $\theta=\tan^{-1}\frac{1.99644}{0.03962}=88.863^\circ$

극점 -25.07924 에서 $\theta=\tan^{-1}\frac{1.99644}{25.11886}=4.544^\circ$

극점 $0.03962-j1.99644$ 에서 $\theta=90^\circ$

$88.863^\circ-4.544^\circ-90^\circ-\theta=-180^\circ$

$\theta=174.319^\circ$

answer

$\theta=185.681^\circ$

- 허수축 통과점

$s^3+25s^2+(100k_t+2)s+100=0$

s^3	1	$100k_t+2$	
s^2	25	100	$25s^2+100=0$ $s^2=\pm j2$
s^1	$100k_t-2$	0	$100k_t-2=0$ $k_t=0.02$
s^0	100		

9-28. 그림 9P-24에 그린 제어시스템의 전방경로 전달함수가 다음과 같다.

$$G(s)=\frac{K(s+\alpha)(s+3)}{s(s^2-1)}$$

(a) $\alpha=5$로 놓고 $K\geq 0$에 대한 근궤적을 그려라.

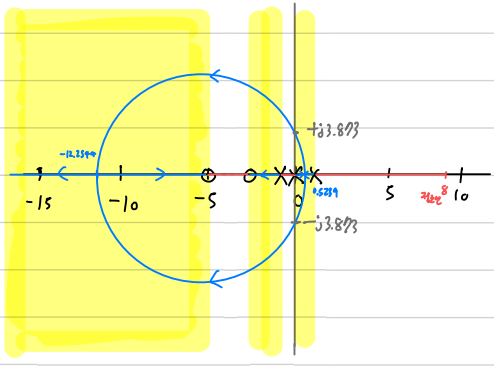
(b) $K=10$으로 놓고 $\alpha\geq 0$에 대한 근궤적을 그려라.

(a) $\alpha=5$, $K\geq 0$

$$1+K\frac{(s+5)(s+3)}{s(s-1)(s+1)}=0$$

영점: $-5,-3$

극점: $-1,0,1$



· 점근선 각도

$n=3$, $m=2$

$\theta=\frac{(2i+1)\times 180}{1}$ $\quad \theta_0=180^\circ$

· 점근선 중심

$$\sigma=\frac{(-1+0+1)-(-5-3)}{3-2}=8$$

· 이탈점, 합류점

$$k=-\frac{s(s^2-1)}{(s+5)(s+3)}=-\frac{s^3-s}{s^2+8s+15} \qquad \frac{dk}{ds}=-\frac{(3s^2-1)(s^2+8s+15)-(s^3-s)(2s+8)}{(s^2+8s+15)^2}=-\frac{s^4+16s^3+46s^2-15}{(s^2+8s+15)^2}$$

$s^4+16s^3+46s^2-15=0$

$s=-12.2544,-3.62489,-0.64458,0.52387$

가능 가능

$s=-12.2544$, $k=27.22856$

$s=0.52387$, $k=0.01953$

· 허수축 통과점

$s^3+ks^2+(8k-1)s+15k=0$

s^3	1	$8k-1$	
s^2	k	$15k$	$ks^2+15k=0$ $\quad s=\pm j3.873$
s^1	$8k-16$		$8k-16=0$ $\quad k=2$
s^0	$15k$		

(b) $K=10,\ \alpha \geq 0$

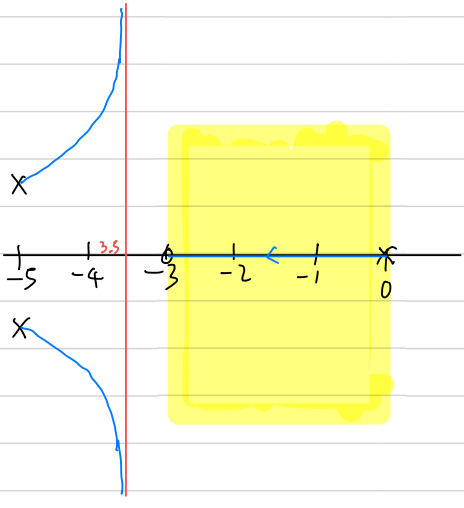
· $K=10, \alpha$에 대한 포르형 만들기

$s^3+10s^2+29s+\alpha(10s+30)=0$

$1+\alpha \dfrac{10(s+3)}{s(s^2+10s+29)}=0$

영점 : -3

극점 : $0, -5\pm j2$



· 점근선

$n=3\ \ m=1$

$\theta=\dfrac{(2i+1)\times 180^\circ}{2}\quad \theta_0=90^\circ,\ \theta_1=270^\circ$

· 근군선 중심

$\sigma=\dfrac{0+(-5+j2)+(-5-j2)-(-3)}{2}=-3.5$

· 이탈점, 합류점

$\alpha=-\dfrac{s^3+10s^2+29s}{10(s+3)}\qquad \dfrac{d\alpha}{ds}=-\dfrac{(3s^2+10s+29)(s+3)-(s^3+10s^2+29s)}{10(s+3)^2}=-\dfrac{2s^3+19s^2+60s+87}{10(s+3)^2}$

$2s^3+19s^2+60s+87=0$

$s=\underbrace{-5.46881}_{\text{불가능}}$, 2개 허수

즉 이탈점, 합류점 X

· 복소극점 출발각

$P=-5+j2$ 기준

영점 -3에서 $\theta=\tan^{-1}\frac{2}{2}=135^\circ$

극점 0에서 $\theta=180^\circ-\tan^{-1}\frac{2}{5}=158.199^\circ$

극점 $-5-j2$에서 $\theta=90^\circ$

$135^\circ-158.199^\circ-90^\circ-\theta=-180^\circ$

$\theta=66.801^\circ$

아래쪽 극점에 대칭이므로

$\theta=293.199^\circ$